

Date-17/03/2026

MIND MATRIX

PROJECT TITLE

HOUSE PRICE PREDICTION

Handwritten Digit Recognition System

4-Day Development Roadmap (4 Team Members)

Team Members & Roles

Member	Role	Responsibility
1.Kowstav Ghosh	Data Engineer	Dataset collection, preprocessing
2.Ritwik Mandal	ML Engineer	Feature extraction and model training
3.Soumadip Tung	Algorithm Engineer	Model optimization & evaluation
4.Debh Sarkar	Integration Engineer	UI / system integration and testing

WEEK-1 : Project Setup & Dataset Preparation

Tasks

Member 1

- Collect dataset (e.g., MNIST)
- Load images using Python
- Explore dataset

Member 2

- Install libraries: NumPy, Pandas, Scikit-learn, OpenCV
- Convert images to grayscale if needed

Member 3

- Perform image preprocessing
 - Noise removal
 - Normalization
 - Resizing

Member 4

- Define system architecture
- Create project folder structure

Output

- Clean dataset ready for training
 - Project environment ready
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WEEK-2 : Feature Extraction & Model Building

Tasks

Member 1

- Prepare training and testing datasets

Member 2

- Extract image features
 - Pixel intensity
 - Edge detection
 - Flatten image arrays

Member 3

- Implement Machine Learning models
 - Logistic Regression
 - Support Vector Machine
 - Neural Network

Member 4

- Build basic Python interface for prediction

Output

- First trained model ready
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WEEK-3 : Model Training & Optimization

Tasks

Member 1

- Train model with training data

Member 2

- Tune hyperparameters

Member 3

- Evaluate model
 - Accuracy
 - Confusion matrix
 - Precision & recall

Member 4

- Connect model with interface

Output

- Optimized trained model
 - Performance evaluation results
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WEEK-4 : Testing, Deployment & Documentation

Tasks

Member 1

- Test with new handwritten images

Member 2

- Improve preprocessing if needed

Member 3

- Final model accuracy verification

Member 4

- Documentation & presentation preparation

Output

- Fully working system
- Final project report
- Demonstration ready

System Workflow (Flowchart)

